Power BI ETL & Data Modeling Project – Summary

Schema Used

A Star Schema was created using Sales_Fact as the main fact table connected with:

- Customer_Dim
- Product_Dim
- Region_Dim
- Date_Dim

Returns_Fact was added separately for return analysis and inactive relationship scenarios.

Relationships Created

- Sales_Fact → Customer_Dim
- Sales_Fact → Product_Dim
- Sales_Fact → Region_Dim
- Sales_Fact → Date_Dim
- Returns_Fact → Sales_Fact
- Returns_Fact → Date_Dim (Inactive)

Single-direction filters were mainly used to avoid ambiguity.

Power Query Work

- Imported all Excel files
- Removed blank rows
- Applied correct data types
- Cleaned and transformed data

Hierarchies Created

- Date: Year → Quarter → Month → Date
- Region: Country → State → City
- Product: Category → Subcategory → ProductName

Verification

Matrix tables were used to verify:

- Sales by Category and Region
- Return Reasons by Fiscal Year
- Revenue by Customer Segment

Issues Resolved

- Fixed data type issues
- Removed blank records
- Resolved ambiguous filters using single-direction relationships
- Handled inactive relationship for returns

Conclusion

The project successfully demonstrates ETL processing, relationship building, hierarchy creation, and Star Schema modeling in Power BI.